

Homework 11

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Problem 1

Consider the metric space $(\mathbb{R}, d)$ with the usual metric $d(x, y) = |x - y|$. Note that the set $\mathbb{Q}$ is not an interval, and so from lecture we know that it must be disconnected.

- (a) Find two sets $A, B \subseteq \mathbb{R}$ that are nonempty and separated so that $\mathbb{Q} = A \cup B$, and prove your answer.
- (b) Find a set $\emptyset \neq C \subsetneq \mathbb{Q}$ that is both open and closed in $\mathbb{Q}$, and prove your answer.

Discussion & Scratch Work

The problem already gives an important fact from lecture: in $(\mathbb{R}, |\cdot|)$, every interval is connected, so if a subset of $\mathbb{R}$ is not an interval, then it cannot be connected. Lecture 29 proves that every interval in $\mathbb{R}$ is connected, and the contrapositive is exactly the idea being invoked here: since $\mathbb{Q}$ is not an interval, it must be disconnected. In Lectures 28 and 29, disconnected means that the set can be written as the union of two nonempty separated sets, where “separated” means

$$\overline{A} \cap B = \emptyset \quad \text{and} \quad A \cap \overline{B} = \emptyset.$$

This tells me what kind of sets I should try to build.

The easiest way to split $\mathbb{Q}$ is to cut at an irrational number, because that creates a genuine gap inside $\mathbb{Q}$. The most natural choice is $\sqrt{2}$. So I define

$$A := (-\infty, \sqrt{2}) \cap \mathbb{Q}, \quad B := (\sqrt{2}, \infty) \cap \mathbb{Q}.$$

These are clearly nonempty, disjoint, and their union is all of $\mathbb{Q}$. The real work is to prove they are separated. The key observation is that, as subsets of $\mathbb{R}$, their closures are

$$\overline{A} = (-\infty, \sqrt{2}], \quad \overline{B} = [\sqrt{2}, \infty),$$

so the only possible boundary point where they might “touch” is $\sqrt{2}$, and that point is not rational. Once that is checked, part (a) is done.

For part (b), the problem is asking for a proper nonempty subset of $\mathbb{Q}$ that is both open and closed in the subspace metric on $\mathbb{Q}$. In plain English, that means a subset of $\mathbb{Q}$ that looks like an open set when viewed from inside $\mathbb{Q}$, but whose complement in $\mathbb{Q}$ also looks open. The same irrational cut works perfectly. If I take

$$C := (-\infty, \sqrt{2}) \cap \mathbb{Q},$$

then it should be open in $\mathbb{Q}$ because it is the intersection of the open set $(-\infty, \sqrt{2})$ in $\mathbb{R}$ with $\mathbb{Q}$, and it should be closed in $\mathbb{Q}$ because it is also the intersection of the closed set $(-\infty, \sqrt{2}]$ in $\mathbb{R}$ with $\mathbb{Q}$. That uses exactly the subspace results from lecture: a set is open or closed in a subspace precisely when it is the intersection of the subspace with a set that is open or closed in the ambient space.

Proof.

(a) Define

$$A := (-\infty, \sqrt{2}) \cap \mathbb{Q}, \quad B := (\sqrt{2}, \infty) \cap \mathbb{Q}.$$

Then $A, B \subseteq \mathbb{R}$, and both are nonempty because, for example,

$$1 \in A, \quad 2 \in B.$$

Also,

$$A \cup B = \mathbb{Q},$$

since every rational number is either less than $\sqrt{2}$ or greater than $\sqrt{2}$, and no rational number equals $\sqrt{2}$ because $\sqrt{2} \notin \mathbb{Q}$.

It remains to show that A and B are separated. First note that

$$\overline{A} = (-\infty, \sqrt{2}] \quad \text{and} \quad \overline{B} = [\sqrt{2}, \infty)$$

in $\mathbb{R}$. Hence

$$\overline{A} \cap B = (-\infty, \sqrt{2}] \cap ((\sqrt{2}, \infty) \cap \mathbb{Q}) = \emptyset,$$

because no number can be both at most $\sqrt{2}$ and strictly greater than $\sqrt{2}$. Similarly,

$$A \cap \overline{B} = ((-\infty, \sqrt{2}) \cap \mathbb{Q}) \cap [\sqrt{2}, \infty) = \emptyset.$$

Therefore A and B are separated.

Since A and B are nonempty, separated, and satisfy $\mathbb{Q} = A \cup B$, this gives a separation of $\mathbb{Q}$.

(b) Let

$$C := (-\infty, \sqrt{2}) \cap \mathbb{Q}.$$

Then $C \neq \emptyset$ since $1 \in C$, and $C \subsetneq \mathbb{Q}$ since, for example, $2 \in \mathbb{Q} \setminus C$.

We show that C is open in $\mathbb{Q}$. The set $(-\infty, \sqrt{2})$ is open in $\mathbb{R}$, so by the subspace topology result from lecture,

$$C = (-\infty, \sqrt{2}) \cap \mathbb{Q}$$

is open in $\mathbb{Q}$.

We next show that C is closed in $\mathbb{Q}$. The set $(-\infty, \sqrt{2}]$ is closed in $\mathbb{R}$, and since $\sqrt{2} \notin \mathbb{Q}$ we have

$$(-\infty, \sqrt{2}] \cap \mathbb{Q} = (-\infty, \sqrt{2}) \cap \mathbb{Q} = C.$$

Again by the subspace result from lecture, this implies that C is closed in $\mathbb{Q}$.

Thus C is a nonempty proper subset of $\mathbb{Q}$ that is both open and closed in $\mathbb{Q}$. □

Problem 2

Let (X, d) be a metric space.

- (a) Prove that if $F \subseteq X$ is disconnected and closed in X , then there exist $A, B \subseteq X$ that are nonempty, separated, and closed in X so that $F = A \cup B$.
- (b) Prove that if $U \subseteq X$ is disconnected and open in X , then there exist $A, B \subseteq X$ that are nonempty, separated, and open in X so that $U = A \cup B$.

Discussion & Scratch Work

The starting point for both parts is the definition of disconnectedness from Lectures 28 and 29: a set is disconnected if it can be written as the union of two nonempty separated sets. And “separated” means

$$\overline{A} \cap B = \emptyset \quad \text{and} \quad A \cap \overline{B} = \emptyset.$$

So in both parts, I do not need to invent the sets from scratch. The definition of disconnectedness already gives me two nonempty separated pieces whose union is the set I started with. The actual work is to prove that those pieces inherit the extra property I want: closedness in part (a), and openness in part (b).

For part (a), I start with $F \subseteq X$ disconnected and closed in X . By disconnectedness, there exist nonempty separated sets $A, B \subseteq F$ with $F = A \cup B$. Since F is closed, I want to use it together with the separation conditions to show that A and B are closed in X . The key observation is that separation forces

$$F \cap \overline{A} = A \quad \text{and} \quad F \cap \overline{B} = B.$$

Once I have that, each set is an intersection of closed sets, so it is closed.

For part (b), I start with $U \subseteq X$ disconnected and open in X . Again, disconnectedness gives nonempty separated sets $A, B \subseteq U$ with $U = A \cup B$. Now I want to prove that A and B are open in X . The right way to use separation is to write each piece as an intersection of U with the complement of the closure of the other piece:

$$A = U \cap (\overline{B})^c, \quad B = U \cap (\overline{A})^c.$$

Since U is open and closures are closed, these are open sets. So the proof in both parts is: use disconnectedness first, and then use either closedness or openness of the ambient set together with the separation conditions to upgrade the two pieces.

Proof.

(a) Suppose that $F \subseteq X$ is disconnected and closed in X . Since F is disconnected, by definition there exist sets $A, B \subseteq X$ such that

$$A \neq \emptyset, \quad B \neq \emptyset, \quad F = A \cup B,$$

and A and B are separated; that is,

$$\overline{A} \cap B = \emptyset \quad \text{and} \quad A \cap \overline{B} = \emptyset.$$

It remains to show that A and B are closed in X .

We claim that

$$F \cap \overline{A} = A.$$

Indeed, since $A \subseteq F$ and $A \subseteq \overline{A}$, we have $A \subseteq F \cap \overline{A}$. Conversely, let $x \in F \cap \overline{A}$. Since $F = A \cup B$, either $x \in A$ or $x \in B$. But $x \in B$ is impossible, because then $x \in \overline{A} \cap B$, contradicting that A and B are separated. Hence $x \in A$. This proves the claim.

Now both F and $\overline{A}$ are closed in X , so $F \cap \overline{A}$ is closed in X . Since $F \cap \overline{A} = A$, it follows that A is closed in X .

Similarly,

$$F \cap \overline{B} = B,$$

and therefore B is closed in X .

Thus there exist $A, B \subseteq X$ that are nonempty, separated, and closed in X such that

$$F = A \cup B.$$

(b) Suppose that $U \subseteq X$ is disconnected and open in X . Since U is disconnected, by definition there exist sets $A, B \subseteq X$ such that

$$A \neq \emptyset, \quad B \neq \emptyset, \quad U = A \cup B,$$

and A and B are separated; that is,

$$\overline{A} \cap B = \emptyset \quad \text{and} \quad A \cap \overline{B} = \emptyset.$$

It remains to show that A and B are open in X .

We claim that

$$A = U \cap (\overline{B})^c.$$

First, let $x \in A$. Then $x \in U$ since $A \subseteq U$, and also $x \notin \overline{B}$ because $A \cap \overline{B} = \emptyset$. Hence

$$x \in U \cap (\overline{B})^c.$$

So $A \subseteq U \cap (\overline{B})^c$.

Conversely, let $x \in U \cap (\overline{B})^c$. Since $x \in U = A \cup B$, either $x \in A$ or $x \in B$. But $x \in B$ is impossible, because every set is contained in its closure, so $x \in B$ would imply $x \in \overline{B}$, contradicting $x \in (\overline{B})^c$. Therefore $x \in A$. This proves the claim.

Since U is open and $(\overline{B})^c$ is open, the set

$$A = U \cap (\overline{B})^c$$

is open in X .

By the same argument,

$$B = U \cap (\overline{A})^c,$$

and therefore B is open in X .

Thus there exist $A, B \subseteq X$ that are nonempty, separated, and open in X such that

$$U = A \cup B.$$

□

Problem 3

Suppose that (X, d) is a metric space such that for any pair of points $a, b \in X$, there exists a connected set $C \subseteq X$ so that $a, b \in C$. Prove that X is connected.

Discussion & Scratch Work

The statement is saying that any two points of X can be placed inside some connected subset of X . Intuitively, that means the space cannot break into two separated pieces, because if it did, then a connected set containing one point from each side would be forced to lie entirely in one side or entirely in the other, which is impossible. So the result I want to prove is really a global connectedness statement coming from the existence of enough connected subsets joining pairs of points.

From Lecture 29, the main result I want is the proposition that if $C \subseteq X$ is connected and $A, B \subseteq X$ are separated with $C \subseteq A \cup B$, then either $C \subseteq A$ or $C \subseteq B$. That is exactly the theorem that prevents a connected set from straddling a separation. From Lectures 28 and 29, I also need the definition of disconnectedness: X is disconnected if there exist nonempty separated sets $A, B \subseteq X$ such that $X = A \cup B$. And of course, X is connected if it is not disconnected.

So the proof template is contradiction. I assume X is disconnected. Then there exist nonempty separated sets A and B whose union is X . I pick one point $a \in A$ and one point $b \in B$. By hypothesis, there is a connected set $C \subseteq X$ containing both a and b . But since $C \subseteq X = A \cup B$, the lecture proposition says that either $C \subseteq A$ or $C \subseteq B$. That cannot happen because C contains both a and b . This contradiction shows that X must be connected.

Proof.

Suppose, toward a contradiction, that X is disconnected. Then by definition there exist sets $A, B \subseteq X$ such that

$$A \neq \emptyset, \quad B \neq \emptyset, \quad X = A \cup B,$$

and A and B are separated; that is,

$$\overline{A} \cap B = \emptyset \quad \text{and} \quad A \cap \overline{B} = \emptyset.$$

Choose points

$$a \in A, \quad b \in B.$$

By the hypothesis of the problem, there exists a connected set $C \subseteq X$ such that

$$a, b \in C.$$

Since $X = A \cup B$, we have

$$C \subseteq A \cup B.$$

Now apply the proposition from lecture: if a connected set is contained in the union of two separated sets, then it must lie entirely in one of them. Therefore either

$$C \subseteq A \quad \text{or} \quad C \subseteq B.$$

But this is impossible, because $a \in C \cap A$ and $b \in C \cap B$. If $C \subseteq A$, then $b \in A$, contradicting that $b \in B$ and A, B are separated (hence disjoint). Similarly, if $C \subseteq B$, then $a \in B$, again a contradiction.

Thus our assumption that X is disconnected is false. Therefore X is connected. $\square$

Problem 4

Let (X, d) be a metric space and $A \subseteq X$. Prove that if A is connected, then $\overline{A}$ is connected.

Discussion & Scratch Work

This problem is really asking me to show that connectedness is preserved when I pass from a set to its closure. The relevant definition from Lectures 28 and 29 is that a set is disconnected if it can be written as the union of two nonempty separated sets; otherwise it is connected. So to prove that $\overline{A}$ is connected, the natural strategy is contradiction: assume that $\overline{A}$ is disconnected, and then try to use that separation to force a separation of A itself.

The key lecture result I want is the proposition from Lecture 29: if C is connected and E, F are separated with $C \subseteq E \cup F$, then either $C \subseteq E$ or $C \subseteq F$. Since A is connected and $A \subseteq \overline{A}$, this proposition says that if $\overline{A}$ were split into two separated pieces, then all of A would have to lie in just one of them. The remaining idea is then to use the fact that points of the closure of A are limits of sequences in A from Lecture 25. If all of A lies in one side of the separation, then every limit point of A must also lie on that same side, which rules out the other side entirely. That contradiction shows $\overline{A}$ cannot be disconnected.

Proof.

Suppose, toward a contradiction, that $\overline{A}$ is disconnected. Then by definition there exist nonempty separated sets $E, F \subseteq X$ such that

$$\overline{A} = E \cup F.$$

Since $A \subseteq \overline{A} = E \cup F$ and A is connected, the proposition from lecture implies that either

$$A \subseteq E \quad \text{or} \quad A \subseteq F.$$

Without loss of generality, assume that

$$A \subseteq E.$$

Now let $x \in F$. Since $F \subseteq \overline{A}$, we have $x \in \overline{A}$. By the sequential characterization of closure from lecture, there exists a sequence $\{a_n\} \subseteq A$ such that

$$a_n \rightarrow x.$$

But every $a_n \in A \subseteq E$, so $\{a_n\}$ is a sequence in E converging to x . Hence

$$x \in \overline{E}.$$

Since also $x \in F$, this shows that

$$x \in \overline{E} \cap F,$$

which contradicts the fact that E and F are separated.

Therefore $F = \emptyset$, contradicting that E and F were both nonempty. This contradiction shows that $\overline{A}$ is not disconnected. Hence $\overline{A}$ is connected. $\square$

Problem 5

Let (X, d) be a metric space and $A, B \subseteq X$. Prove that if A and B are both connected and $A \cap B \neq \emptyset$, then $A \cup B$ is connected.

Discussion & Scratch Work

This is one of the standard closure properties of connected sets. The main definition from Lectures 28 and 29 is that a set is disconnected if it can be written as the union of two nonempty separated sets; otherwise it is connected. So the natural proof strategy is contradiction: assume that $A \cup B$ is disconnected, and then analyze what that separation would force on the connected sets A and B individually.

The key lecture result I need is the proposition from Lecture 29 that says: if C is connected and E, F are separated with $C \subseteq E \cup F$, then either $C \subseteq E$ or $C \subseteq F$. That theorem is exactly designed for this problem. If $A \cup B$ were disconnected, then I could write

$$A \cup B = E \cup F$$

with E and F nonempty and separated. Since A is connected and lies in $E \cup F$, the proposition forces all of A to lie in one side. Likewise, all of B must lie in one side. Because $A \cap B \neq \emptyset$, the sets A and B cannot land in different sides. So they must both lie in the same side, which would force the other side to be empty. That contradiction shows that $A \cup B$ is connected.

Proof.

Suppose, toward a contradiction, that $A \cup B$ is disconnected. Then by definition there exist nonempty separated sets $E, F \subseteq X$ such that

$$A \cup B = E \cup F.$$

Since $A \subseteq A \cup B = E \cup F$ and A is connected, the proposition from lecture implies that either

$$A \subseteq E \quad \text{or} \quad A \subseteq F.$$

Similarly, since $B \subseteq E \cup F$ and B is connected, either

$$B \subseteq E \quad \text{or} \quad B \subseteq F.$$

We claim that A and B must lie in the same side. Indeed, suppose not. Then, after relabeling if necessary, we would have

$$A \subseteq E \quad \text{and} \quad B \subseteq F.$$

But $A \cap B \neq \emptyset$, so choose some point $x \in A \cap B$. Then $x \in E$ and $x \in F$, which implies

$$x \in E \cap F.$$

This is impossible because separated sets are disjoint. Hence A and B must both be contained in E , or both be contained in F .

If $A \subseteq E$ and $B \subseteq E$, then

$$A \cup B \subseteq E,$$

so from $A \cup B = E \cup F$ we get $F = \emptyset$, contradicting that F is nonempty. The case $A \subseteq F$ and $B \subseteq F$ similarly implies $E = \emptyset$, also a contradiction.

Thus our assumption that $A \cup B$ is disconnected is false. Therefore $A \cup B$ is connected. $\square$

Problem 6

Let (X, d) be a metric space.

- (a) Let $A, B, C \subseteq X$. Prove that if the sets A and B are separated and the sets A and C are separated, then the sets A and $B \cup C$ are separated.
- (b) Prove that if $F, G \subseteq X$ are closed and $F \cup G$ and $F \cap G$ are connected, then F is connected.

Discussion & Scratch Work

For part (a), the whole problem is about carefully unpacking the definition of separated sets. From lecture, E and F are separated if

$$\overline{E} \cap F = \emptyset \quad \text{and} \quad E \cap \overline{F} = \emptyset.$$

So if I want to prove that A and $B \cup C$ are separated, I need to check exactly those two conditions with $B \cup C$ in place of one of the sets. The first condition is easy, because

$$\overline{A} \cap (B \cup C) = (\overline{A} \cap B) \cup (\overline{A} \cap C),$$

and both pieces are empty by hypothesis. For the second condition, the key topological fact is

$$\overline{B \cup C} \subseteq \overline{B} \cup \overline{C}.$$

Then

$$A \cap \overline{B \cup C} \subseteq (A \cap \overline{B}) \cup (A \cap \overline{C}),$$

and again both pieces are empty. So part (a) is really just a direct closure computation.

For part (b), the assumptions are that F and G are closed, that $F \cup G$ is connected, and that $F \cap G$ is connected. I want to prove that F itself is connected. The natural strategy is contradiction: assume F is disconnected. Then by Problem 2(a), because F is also closed, I can write

$$F = A \cup B$$

where A and B are nonempty, separated, and closed in X . Since $F \cap G$ is connected and sits inside $A \cup B$, the connected-set proposition from Lecture 29 says that $F \cap G$ must lie entirely in one side. Without loss of generality, that means

$$F \cap G \subseteq A.$$

Then $G \cap B = \emptyset$. Since both G and B are closed, that actually makes G and B separated. We already know A and B are separated, so by part (a), B is separated from $A \cup G$. But then

$$F \cup G = (A \cup G) \cup B$$

is a separation of $F \cup G$, contradicting connectedness. So F must be connected.

Proof.

(a) Suppose that A and B are separated and that A and C are separated. We show that A and $B \cup C$ are separated.

First,

$$\overline{A} \cap (B \cup C) = (\overline{A} \cap B) \cup (\overline{A} \cap C).$$

Since A and B are separated, we have $\overline{A} \cap B = \emptyset$, and since A and C are separated, we have $\overline{A} \cap C = \emptyset$. Therefore

$$\overline{A} \cap (B \cup C) = \emptyset.$$

Next, because $B \subseteq B \cup C$ and $C \subseteq B \cup C$, we have

$$\overline{B \cup C} \subseteq \overline{B} \cup \overline{C}.$$

Hence

$$A \cap \overline{B \cup C} \subseteq A \cap (\overline{B} \cup \overline{C}) = (A \cap \overline{B}) \cup (A \cap \overline{C}).$$

Again, since A is separated from both B and C , we have

$$A \cap \overline{B} = \emptyset \quad \text{and} \quad A \cap \overline{C} = \emptyset.$$

Therefore

$$A \cap \overline{B \cup C} = \emptyset.$$

Thus

$$\overline{A} \cap (B \cup C) = \emptyset \quad \text{and} \quad A \cap \overline{B \cup C} = \emptyset,$$

so A and $B \cup C$ are separated.

(b) Suppose, toward a contradiction, that F is disconnected. Since F is also closed in X , Problem 2(a) implies that there exist sets $A, B \subseteq X$ such that

$$A \neq \emptyset, \quad B \neq \emptyset, \quad F = A \cup B,$$

and A and B are separated and closed in X .

Now consider the set $F \cap G$. Since $F \cap G \subseteq F = A \cup B$ and $F \cap G$ is connected by assumption, the proposition from Lecture 29 implies that either

$$F \cap G \subseteq A \quad \text{or} \quad F \cap G \subseteq B.$$

Without loss of generality, assume that

$$F \cap G \subseteq A.$$

We claim that G and B are separated. Indeed, since $B \subseteq F$, we have

$$G \cap B \subseteq G \cap F = F \cap G \subseteq A.$$

But A and B are separated, hence disjoint, so $A \cap B = \emptyset$. Therefore

$$G \cap B = \emptyset.$$

Because G and B are both closed in X , we have $\overline{G} = G$ and $\overline{B} = B$, and so

$$\overline{G} \cap B = G \cap B = \emptyset, \quad G \cap \overline{B} = G \cap B = \emptyset.$$

Thus G and B are separated.

We already know that A and B are separated. Applying part (a) with the roles of the sets relabeled, we conclude that B and $A \cup G$ are separated.

Now

$$F \cup G = (A \cup B) \cup G = (A \cup G) \cup B.$$

Also, $B \neq \emptyset$ by construction, and $A \cup G \neq \emptyset$ because $A \neq \emptyset$. Since B and $A \cup G$ are separated, this shows that $F \cup G$ is disconnected, contradicting the assumption that $F \cup G$ is connected.

Therefore F must be connected. □

Appendix: Toolbox